

Python Basics – Day2

1. Variables in Python

Variables in Python are used to store data values. They are created when you assign a value to them, and you don't need to declare their type (Python is dynamically typed).

Syntax for Variable Assignment:

```
x = 5 # Assigning an integer value to the variable x
```

```
y = "Hello" # Assigning a string value to the variable y
```

Variable Naming Rules:

- Variable names can contain letters (a-z, A-Z), numbers (0-9), and underscores (_).
- Variable names must start with a letter or an underscore.
- Variable names are case-sensitive (Name and name are different).

Example:

```
age = 25
```

```
name = "John"
```

```
is_student = True
```

2. Data Types in Python

Python has various built-in data types. Some common ones are:

- **int**: For integers (e.g., 1, -3, 100)
- **float**: For floating-point numbers (e.g., 3.14, -0.001)
- **str**: For strings (e.g., "Hello", "Python")
- **bool**: For boolean values (True or False)

Type Checking:

You can use the `type()` function to check the type of a variable.

```
x = 10
```

```
print(type(x))
```

3. Type Conversion

Python allows you to convert between data types using functions like `int()`, `float()`, `str()`, etc.

Example:

```
x = "10" # x is a string
```

```
y = int(x) # Convert string to integer
```

```
z = float(y) # Convert integer to float
```

```
print(z)
```

4. Arithmetic Operators

Python supports basic arithmetic operations like addition, subtraction, multiplication, division, and more.

Common Operators:

- + (Addition)
- - (Subtraction)
- * (Multiplication)
- / (Division)
- // (Floor Division)
- % (Modulus)
- ** (Exponentiation)

Examples:

```
a = 10
```

```
b = 3
```

```
print(a + b)
```

```
print(a - b)
```

```
print(a * b)
```

```
print(a / b)
```

```
print(a // b)
```

```
print(a % b)
```

```
print(a ** b)
```